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by

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Ottawa, May 1983

NATIONAL RESEARCH COUNCIL CANADA

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INTRODUCTION

This report presents the results of a series of measurements undertaken in the DBR-NRC reverberation room to determine whether the room meets both the broad-band and the discrete-frequency qualification requirements of the American National Standard for sound power measurements.[1,2]

Our results give additional support to other researchers' findings that loudspeakers with a diameter larger than 200 mm are not suitable for discrete-frequency tests at high frequencies and that averaging over source positions is not very effective at low frequencies.

For the broad-band requirements, it is possible to qualify the DBR room with its present configuration of fixed diffusers and a rotating vane. This is also true for the discrete-frequency requirement for practically any single source position except for the two lowest frequency bands (100 Hz and 125 Hz). To qualify the room for these two bands, four resonant type low frequency absorbers are required and it is necessary to average over at least two specifically chosen source locations.

Since the DBR room configuration is still under continuous improvement for other purposes, we have not optimized the amount of absorption to qualify the room for any single source position. The main purpose of this report is to summarize the steps involved in the qualification procedure and to point out beneficial modifications that might be useful for future qualification tests.

FACILITY DESCRIPTION

The reverberation room to be qualified for sound power measurement is the larger of the two rooms comprising the transmission loss suite in the Division of Building Research. It is rectangular, with a nominal volume of 255 m^3 and dimensions of $8.0 \times 6.5 \times 4.9$ m. In its general use, the room is equipped with fixed diffusers and a rotating vane. The fixed diffusers are made of 12.7 mm thick sheet plywood, 1.21×1.21 m. Nine of these are hung at random in space. An additional panel, 0.9×2.2 m, is placed along one of the lower edges of the room to improve diffusion in the lower portion of the room. The rotating vane consists of four 1.21×2.42 m sheets of 12.7 mm thick plywood arranged symmetrically about the axis of rotation. The vane is mounted in the middle of the upper portion of the room and rotates at a rate of 12 rpm.

The room has automatic environmental controls and is generally maintained at 21°C and 55% relative humidity.

INSTRUMENTATION

The existing instrumentation system in the Noise and Vibration Section for commercial testing is quite readily adapted for these qualification tests. Generation of the sequence of test tones, and acquisition and numerical processing of the data can be performed automatically under computer control. A block diagram of this system is shown in Fig. 1.

An array of nine GR Type 1961-9602 25.4 mm electret microphones is placed in the reverberation room according to the standard's requirement. These microphones are mounted on GR Type 1560-P42 preamplifiers, which are in turn connected to the GR Multichannel Amplifier and Multiplexer. Under computer control, any one of the microphones can be selected and its output fed into the GR Real-Time Analyzer. The latter consists of a Type 1925 Multifilter and a Type 1926 Multichannel RMS Detector. The RMS Detector computes the root-mean-square voltage levels, referenced to 1 mV, from the digitized samples obtained for each one-third octave band over the integration period. These data are transmitted to the computer, which controls the integration time and starts the data transfer process.

The qualification test tones are generated by a Rockland Frequency Synthesizer. The signal frequency and amplitude from this synthesizer are also controlled by the computer.

ACOUSTIC SOURCES

For the broad-band qualification test, an ILG Reference Sound Source was used as the noise generator. Two loudspeakers of different dimensions were used for the pure-tone qualification test. One was a 120 mm KEF B110 loudspeaker mounted in a closed cubic box 23 cm on each side. The other was a 250 mm JBL E110 loudspeaker mounted in a closed rectangular box measuring 30 × 35 × 25 cm. The loudspeaker responses were measured in an anechoic room over a reflecting plane with the speaker cone facing upward in accordance with the standard.

In performing these measurements, another GR microphone of the same type was used. It was placed about 10 mm above the plane of the speaker baffle on the speaker axis with the microphone diaphragm perpendicular to the speaker baffle so that sound from the speaker struck it at grazing incidence. According to the manufacturer, the frequency response of this type of microphone is practically the same at both grazing and random incidence up to 2.5 kHz. This ensures that the microphones measuring the near-field response have the same response as those measuring the reverberant field. As shown in Tables I and II, the near-field responses of both loudspeakers satisfied the standard's requirement that sound pressure levels at adjacent frequencies do not differ by more than 1 dB.

ROOM QUALIFICATION

The qualification tests were performed according to the procedure described in the two American National Standards, ANSI S1.31-1980 and ANSI S1.32-1980.[1,2]

Broad-Band Test

For the broad-band case, the procedure is fairly simple. Sound pressure levels at the nine microphone positions were measured for eight source locations using the ILG as the noise source. The relative source locations are shown in Fig. 2

Since spatial standard deviation is required for this test, all the microphones have to be calibrated before the experiment. The calibration was performed with a B & K Type 4230 Sound Level Calibrator at 1 kHz. It has an accuracy of ±0.3 dB, as required by the standard.

For each frequency band for which the test room is to be qualified, the spatial standard deviation (S_s) in decibels, was computed using the formula:

$$S_s = \{\frac{1}{(N_s - 1)} \sum_{j=1}^{N_s} [(L_m)_j - \langle L_m \rangle]^2\}^{\frac{1}{2}} \tag{1}$$

where: $(L_m)_j$ = sound pressure level averaged over all microphone positions when the ILG is in the jth source location, (dB);

$\langle L_m \rangle$ = arithmetic mean of $(L_m)_j$ values, averaged over all source locations (dB); and,

N_s = number of reference sound source positions (eight were used).

The test room will qualify for the measurement of broad-band noise sources if the computed standard deviation does not exceed the limits specified in the standard. Results presented in Table III show that the existing DBR room, with fixed diffusers and rotating vane, qualified for all the frequency bands from 100 Hz to 10 kHz.

Pure-Tone Test

From a practical point of view, it is important to know the effects of various room configurations and measurement parameters on the results of the qualification tests. The main objective is to find a room configuration such that the room can be qualified for a single source position placed anywhere within a certain area in the room. With this in mind, we have carried out tests in the large DBR-NRC reverberation chamber for three different configurations. The alternative procedure for the measurement of discrete-frequency components as set forth in Section 5 of ANSI S1.32-1980[2] was followed.

For each one-third octave band, the sound pressure levels were measured at nine independent microphone positions at the prescribed test frequencies listed in Table V of ANSI S1.32-1980, using one of the loudspeakers as the source. The near-field characteristic of the loudspeaker was then corrected for at each frequency before the arithmetic mean, $\langle L_p \rangle$, and the standard deviation, S_f, were computed using the following equation:

$$S_f = \left\{ \frac{1}{(n-1)} \sum_{k=1}^{n} \left[(L_p)_k - \langle L_p \rangle \right]^2 \right\}^{\frac{1}{2}} \qquad (2)$$

where: $(L_p)_k$ = average sound pressure level (corrected for loudspeaker response) produced in the test room by the loudspeaker source when excited at the kth test frequency, average over all microphone positions (and if appropriate, over all loudspeaker source locations) (dB);

$\langle L_p \rangle$ = arithmetic mean of $(L_p)_k$ values, averaged over all n test frequencies (dB);

n = number of test frequencies in a given one-third octave band.

The room will qualify for measurements of narrow-band or discrete-frequency noise sources if the computed standard deviations do not exceed the limits specified in the standard. Results and discussions of these tests will be presented in the following sections.

Room with fixed diffusers - For this case, the rotating vane was not in operation but was fixed at a particular orientation for the complete set of experiments. Six source positions have been tested, using the JBL loudspeaker. Locations of these source positions are the same as those shown in Fig. 2.

Results presented in Table IV show that the room in this configuration did not qualify for any one of the source locations used. Even the averaged results over the six source positions did not qualify the room at the lower frequency bands. The reason for the ineffectiveness in source position averaging has been given by Ebbing and Maling.[3] They suggest that at the lower frequencies, because the modal overlap is not sufficiently great, the sound power output of the source at a particular frequency may be high (or low) regardless of the position of the source in the room. Thus, plots of space-averaged sound pressure levels in the room versus frequency for several source positions are highly correlated, as shown in Fig. 13 of their report.[3]

Similar results were obtained by Lang and Rennie.[4] Our present results show the behaviour at the lower frequencies depicted in Fig. 3.

The relatively poor performance at the high frequency bands, as indicated in Table IV, was found to be a consequence of the large JBL loudspeaker being used. Further discussion on this will be given in the next section.

Room with fixed diffusers and a rotating vane - Significant improvement was obtained in general when the rotating vane was in operation, as indicated by results tabulated in Table V. The room in this configuration qualified for almost any source position tested, except at the two lowest frequency bands. The larger standard deviations measured at the high frequency bands were due to the use of the large JBL loudspeaker. More acceptable values were obtained when the JBL loudspeaker was replaced by the smaller KEF loudspeaker, as shown in Table V. The anomaly might be caused by the directionality of the loudspeaker. Fig. 4 compared the directivities of the two loudspeakers measured in an anechoic chamber. The JBL loudspeaker shows a more severe directional characteristic than the KEF loudspeaker in the 2.5 kHz band. Unfortunately, the latter does not have enough acoustic output at low frequencies to be used for the full frequency range of the tests.

When averaging over source position was applied to the data, it was found that the room in this configuration still failed to qualify at the 100 Hz one-third octave band even when seven source positions were used. In fact, plots of the space-averaged sound pressure level in the room versus frequency show that the rotating vane did not produce significant changes in the shape of the response curves from those of the fixed diffusers at this lowest frequency band. Typical curves for two different source locations are shown in Figs. 5 and 6. It is not clear at this point whether it is the size or the paddle design of the rotating vane that is the source of the problem. Another possible explanation is that the ratio of room dimension to wavelength has approached a limit that makes it difficult to change the modal structure of the room. For the 100 Hz band, the modal overlap of the room is about 0.4. The next logical step is to add absorption to broaden the modal band-width and smooth out the response curves. Detailed discussion will be given in the next section.

Room with fixed diffusers, rotating vane and low-frequency absorbers - In order to provide absorption only at the lowest frequency bands, some form of resonant type absorbers has to be used. The one chosen consisted of a rectangular wooden box 61 × 122 × 15 cm. Ordinary pegboard was used for the top face with alternate rows and columns of the holes taped. This left a spacing of about 7.6 cm between holes to provide a resonance frequency at about 120 Hz. The box was filled with low density fiberglass. Four of these were used, two hung on the side walls and two placed along the edges of the room. These absorbers provide fairly good absorptions at low frequencies, as indicated by the changes in reverberation time presented in Fig. 7. The smoothing effect on the room responses is also evident from the plots shown in Figs. 5 and 6.

Although further improvement to the measured standard deviation has been obtained, it was still not sufficient to qualify the room at any single source position, as indicated by results shown in Table VI. We are, however, approaching the possibility of qualifying the room with averaging results over any pair of source positions, as suggested by results shown in Table VII.

With the low frequency absorbers added, the average Sabin absorption coefficient of the room at 100 Hz was about 0.05, quite a bit below the value of 0.16 allowed by the standard. Thus, we are optimistic that the room can qualify even for any single source position if more low frequency absorption is added. These additional investigations will be performed when we have perfected an alternate procedure[5] that is faster and more accurate.

No data above 1 kHz for this room configuration has been presented because we are confident the room will qualify for these frequencies if the smaller KEF loudspeaker is used.

CONCLUSIONS

The basic conclusions derived from this study are as follows: (1) loudspeakers with diameter larger than 200 mm are not suitable for the discrete-frequency qualification test at high frequencies; (2) at low frequencies, the space-averaged room response shows significant correlation over different source locations, thus rendering averaging over source positions relatively ineffective; (3) to qualify the existing DBR-NRC reverberation room for discrete-frequency measurement over the whole frequency range, it is necessary to add low frequency absorption and use two source positions; (4) qualification of the room for broad-band measurement is possible without the addition of low frequency absorptions.

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TABLE I

NEAR-FIELD RESPONSE OF THE JBL LOUDSPEAKER

Center Frequency of One-Third Octave Bands (Hz)														
100	125	160	200	250	315	400	500	630	800	1000	1250	1600	2000	2500
		99.13				101.75						102.00		
	96.75	99.25		99.75		101.75					102.00	102.06		95.50
	97.06	99.50		100.00		101.75	101.63	100.13	102.25		102.00	102.06		95.50
94.25	97.25	99.50	99.56	100.00	100.00	102.00	101.50	100.06	102.69	101.13	102.19	102.06	99.32	95.63
94.50	97.50	99.50	99.75	100.00	100.44	102.00	101.81	100.13	102.81	101.69	102.38	102.00	99.63	95.81
94.75	97.75	99.75	100.00	100.25	100.75	102.00	101.76	100.32	103.00	102.19	102.50	102.00	99.94	96.06
95.00	98.00	100.00	100.25	100.25	101.00	102.25	101.81	100.50	103.25	102.38	102.63	101.94	100.00	96.44
95.25	98.00	100.00	100.38	100.50	101.25	102.25	101.94	100.56	103.50	102.69	102.81	101.88	100.25	96.88
95.50	98.25	100.00	100.50	100.50	101.50	102.25	102.00	100.82	103.50	102.69	102.94	101.50	100.50	97.00
95.50	98.25	100.00	100.50	100.50	101.50	102.25	102.25	100.94	103.50	102.88	103.00	101.50	100.69	97.31
95.75	98.50	100.19	100.75	100.50	101.75	102.50	102.13	101.13	103.25	103.00	103.25	101.31	100.75	97.81
96.00	98.50	100.13	100.75	100.50	101.88	102.44	102.06	101.13	103.25	103.00	103.13	101.25	101.00	98.07
96.25	98.50	100.25	100.75	100.50	102.00	102.50	102.13	101.38	103.00	103.06	103.19	101.19	100.75	98.31
96.50	98.75	100.25	100.81	100.75	102.00	102.50	102.06	101.57	102.81	103.00	103.50	101.00	100.76	98.63
96.50	98.75	100.25	100.75	100.50	102.00	102.50	102.13	101.69	102.56	103.00	103.69	100.88	100.75	98.81
96.56	98.75	100.25	100.75	100.69	102.00	102.50	101.81	102.00	102.31	103.06	103.63	100.63	100.75	99.00
96.75	99.00	100.25	100.75	100.69	102.00	102.56	101.82	102.06	102.06	103.00	103.88	100.50	100.88	98.88
96.81	99.00	100.25	100.75	100.56	102.06	102.50	101.76	102.25	101.63	102.88	104.00	100.25	100.63	98.94
97.00	99.00	100.25	100.75	100.50	102.06	102.50	101.63	102.44	101.50	102.88	104.13	100.06	100.38	99.00
97.00	99.00	100.25	100.56	100.56	102.00	102.38	101.44	102.50	101.25	102.81	104.06	99.81	100.19	98.75
97.00	99.00	100.25	100.50	100.50	102.00	102.38	101.25	102.69	101.06	102.56	104.06	99.50	99.50	98.94
97.06	99.00	100.25	100.50	100.50	102.00	102.31	101.19	102.56	101.00	102.50	104.13	99.38	98.94	98.76
97.13	99.00	100.25	100.25	100.50	102.00	102.00	101.00	102.75	100.88	102.38	103.88	99.32	98.06	98.50
97.00	99.00	100.25	100.00	100.50	101.75	102.25	100.88	102.69	100.88	102.38	103.56	98.94	97.38	98.50
97.00	99.00	100.19	100.00	100.25	101.75	102.19	100.63	102.50	101.00	102.25	103.50	99.00	96.75	98.13
	99.00	100.00		100.25		102.06		102.50			103.13	98.69		98.00
	98.75	100.00		100.06		101.88					103.00	98.56		98.00

TABLE II

NEAR-FIELD RESPONSE OF THE KEF LOUDSPEAKER

Center Frequency of One-Third Octave Bands (Hz)							
500	630	800	1000	1250	1600	2000	2500
					102.00		
				103.25	102.13		100.50
104.38	103.88	104.75		103.00	102.25		100.63
104.19	103.88	105.00	105.88	102.88	102.25	102.63	100.63
104.25	103.75	105.25	106.13	102.50	102.31	102.75	100.69
104.38	104.19	105.50	106.13	102.13	102.50	102.63	100.75
104.63	104.25	105.75	106.19	101.94	102.88	102.63	100.75
104.63	104.44	106.00	106.13	101.38	103.50	102.63	100.81
104.76	104.63	106.25	106.25	101.06	104.38	102.44	100.81
104.88	104.63	106.25	106.13	100.50	105.13	102.38	100.75
104.88	105.00	106.50	106.13	100.31	105.25	102.31	100.69
105.00	105.06	106.50	106.32	100.07	105.13	102.25	100.63
104.81	105.13	106.56	106.25	99.63	104.81	102.25	100.63
105.00	105.00	106.50	106.06	99.38	104.50	102.13	100.50
104.81	105.25	106.82	106.19	99.31	104.25	102.00	100.38
104.88	105.38	106.75	106.13	99.00	104.25	102.13	100.31
104.75	105.38	106.56	106.00	99.01	104.00	101.82	100.38
104.94	105.38	106.88	105.88	99.13	104.00	101.63	100.19
104.74	105.25	106.63	105.56	99.06	103.69	101.69	100.06
104.50	105.31	106.63	105.50	99.07	103.81	101.44	100.19
104.38	105.38	106.50	105.31	99.25	103.69	101.32	100.13
104.25	105.38	106.44	105.06	99.19	103.50	101.13	100.00
104.25	105.19	106.19	104.81	99.19	103.50	101.00	99.81
104.00	105.19	106.13	104.38	99.06	103.50	100.81	99.88
103.94	105.00	106.00	104.00	99.13	103.25	100.75	99.75
	105.00			99.13	103.25		99.63
				99.13	103.25		99.50

TABLE III

BROAD-BAND QUALIFICATION DATA FOR ROOM WITH FIXED DIFFUSERS AND A ROTATING VANE

One-Third Octave Band Center Frequency (Hz)	Maximum Allowable Standard Deviations (dB)	Measured Standard Deviations (dB)
100	1.5	0.5
125	1.5	0.5
160	1.5	0.3
200	1.0	0.2
250	1.0	0.2
315	1.0	0.3
400	1.0	0.2
500	1.0	0.2
630	1.0	0.2
800	0.5	0.1
1000	0.5	0.2
1250	0.5	0.1
1600	0.5	0.2
2000	0.5	0.2
2500	0.5	0.2
3150	1.0	0.2
4000	1.0	0.2
5000	1.0	0.2
6300	1.0	0.3
8000	1.0	0.5
10000	1.0	0.5

TABLE IV

DISCRETE-FREQUENCY QUALIFICATION DATA FOR ROOM WITH FIXED DIFFUSERS

One Third Octave Band Center Frequency	Maximum Allowable Standard Deviations	Measured Standard Deviations At Difference Source Positions #1	#2	#3	#4	#5	#6	Averaged <1-6>
100	3	5.6	4.9	3.9	4.8	5.5	4.9	4.3
125	3	4.4	6.1	4.2	4.9	4.9	3.8	3.8
160	3	4.7	5.2	3.7	4.4	3.4	3.6	3.0
200	2	2.9	2.8	3.5	2.8	3.3	4.0	2.2
250	2	3.6	3.0	3.8	3.3	3.7	3.3	2.1
315	2	3.1	2.0	2.9	2.9	2.7	3.2	1.3
400	1.5	2.3	2.8	1.7	2.5	2.8	2.3	1.2
500	1.5	1.9	1.7	2.5	2.9	2.1	2.5	1.2
630	1.5	1.8	2.0	1.6	2.3	1.8	2.0	0.7
800	1	1.9	2.0	1.7	1.4	1.2	1.4	0.8
1000	1	1.8	1.8	2.2	1.6	2.3	1.5	1.0
1250	1	1.8	2.1	1.7	1.6	1.6	1.9	1.1
1600	1	1.9	1.8	2.0	1.7	2.1	1.8	1.2
2000	1	2.3	1.6	1.9	1.8	1.9	1.7	1.2
2500	1	2.0	1.9	2.5	2.3	2.0	2.4	1.8

TABLE V

DISCRETE-FREQUENCY QUALIFICATION DATA FOR ROOM WITH FIXED DIFFUSERS AND A ROTATING VANE

	One-Third Octave Band Center Frequency	Maximum Allowable Standard Deviations	Measured Standard Deviations At Different Source Positions #1	#2	#3	#4	#6	#7	#8	Averaged <1-8>
Using 250 mm JBL Speaker	100	3	4.3	4.2	3.4	4.3	4.4	4.0	4.7	3.1
	125	3	3.8	4.8	3.5	4.3	3.2	3.6	3.0	2.7
	160	3	2.4	3.7	2.3	2.7	2.2	2.9	2.3	1.6
	200	2	1.7	1.9	1.9	1.8	2.1	1.8	1.5	1.1
	250	2	2.1	1.1	1.7	1.8	1.2	1.6	1.9	0.7
	315	2	1.1	1.2	1.1	1.1	1.4	1.4	1.1	0.5
	400	1.5	0.9	1.0	1.1	1.1	0.9	0.8	1.0	0.5
	500	1.5	0.6	1.1	0.9	0.9	0.6	0.9	0.7	0.4
	630	1.5	0.6	0.6	0.6	0.6	0.6	0.4	0.7	0.3
	800	1	0.7	0.4	0.6	0.6	0.5	0.6	0.4	0.3
	1000	1	0.8	0.7	0.7	0.7	0.6	0.7	0.6	0.5
	1250	1	0.9	1.0	0.7	1.0	0.6	0.8	0.7	0.7
	1600	1	1.0	1.1	1.1	1.0	0.9	1.0	0.9	0.9
	2000	1	1.1	0.9	1.0	1.2	1.1	0.9	0.8	0.9
	2500	1	1.6	1.6	1.8	1.6	1.6	1.8	1.7	1.6
Using 120 mm KEF Speaker	1000	1		0.6		0.8	0.6		0.7	
	1250	1		0.8		0.6	0.6		0.7	
	1600	1		0.7		1.0	0.7		0.8	
	2000	1		0.4		0.4	0.4		0.4	
	2500	1		0.6		0.8	0.6		0.5	

TABLE VI

DISCRETE-FREQUENCY QUALIFICATION DATA FOR ROOM WITH FIXED DIFFUSERS, ROTATING VANE AND LOW FREQUENCY ABSORBERS

One-Third Octave Band Center Frequency	Maximum Allowable Standard Deviation	Measured Standard Deviations At Different Source Positions							
		#1	#2	#3	#4	#5	#6	#7	#8
100	3	4.1	3.3	3.2	2.8	3.8	3.8	3.4	4.0
125	3	3.5	4.7	2.7	2.9	3.3	2.5	2.5	2.6
160	3	2.5	2.6	2.3	2.1	1.8	2.4	3.0	2.0
200	2	1.7	1.6	1.6	2.2	2.2	1.9	1.5	1.7
250	2	1.9	1.3	1.5	1.0	1.5	1.6	1.6	1.6
315	2	1.3	1.0	1.0	1.1	1.1	1.4	1.3	1.0
400	1.5	.8	1.1	1.1	1.0	0.7	1.0	1.0	1.1
500	1.5	.9	1.0	0.8	0.9	0.8	0.8	1.1	0.6
630	1.5	.7	0.5	0.7	0.8	0.8	0.7	0.6	0.7
800	1.0	.7	0.6	0.6	0.6	0.7	0.7	0.6	0.5
1000	1.0	.8	0.6	0.8	0.7	0.6	0.8	0.7	0.5

TABLE VII

AVERAGED RESULTS OF TABLE VI

One-Third Octave Band Center Frequency	Maximum Allowable Standard Deviation	Averaged Standard Deviations of Different Pairs of Source Positions							
		<2,7>	<3,4>	<4,7>	<2,3>	<3,7>	<2,4>	<5,4>	<6.4>
100	3	2.9	2.7	3.0	2.9	3.1	2.6	3.1	3.2
125	3	2.7	2.7	2.3	3.6	2.1	3.1	2.8	2.4
160	3	2.3	2.2	2.9	2.2	2.4	2.5	1.7	2.3
200	2	1.1	1.5	1.5	1.3	1.3	1.5	2.2	1.8
250	2	1.2	1.3	1.6	1.2	1.5	1.3	1.5	1.6
315	2	1.0	0.9	1.3	1.0	1.2	1.0	0.7	1.4
400	1.5	0.8	1.0	1.0	1.0	0.9	1.0	0.6	1.0
500	1.5	0.8	0.8	1.1	0.9	1.0	1.0	0.7	0.8
630	1.5	0.4	0.6	.06	0.4	0.6	0.5	0.6	0.7
800	1.0	0.5	0.6	0.6	0.6	0.5	0.6	0.5	0.7
1000	1.0	0.6	0.8	0.7	0.6	0.6	0.6	0.6	0.8

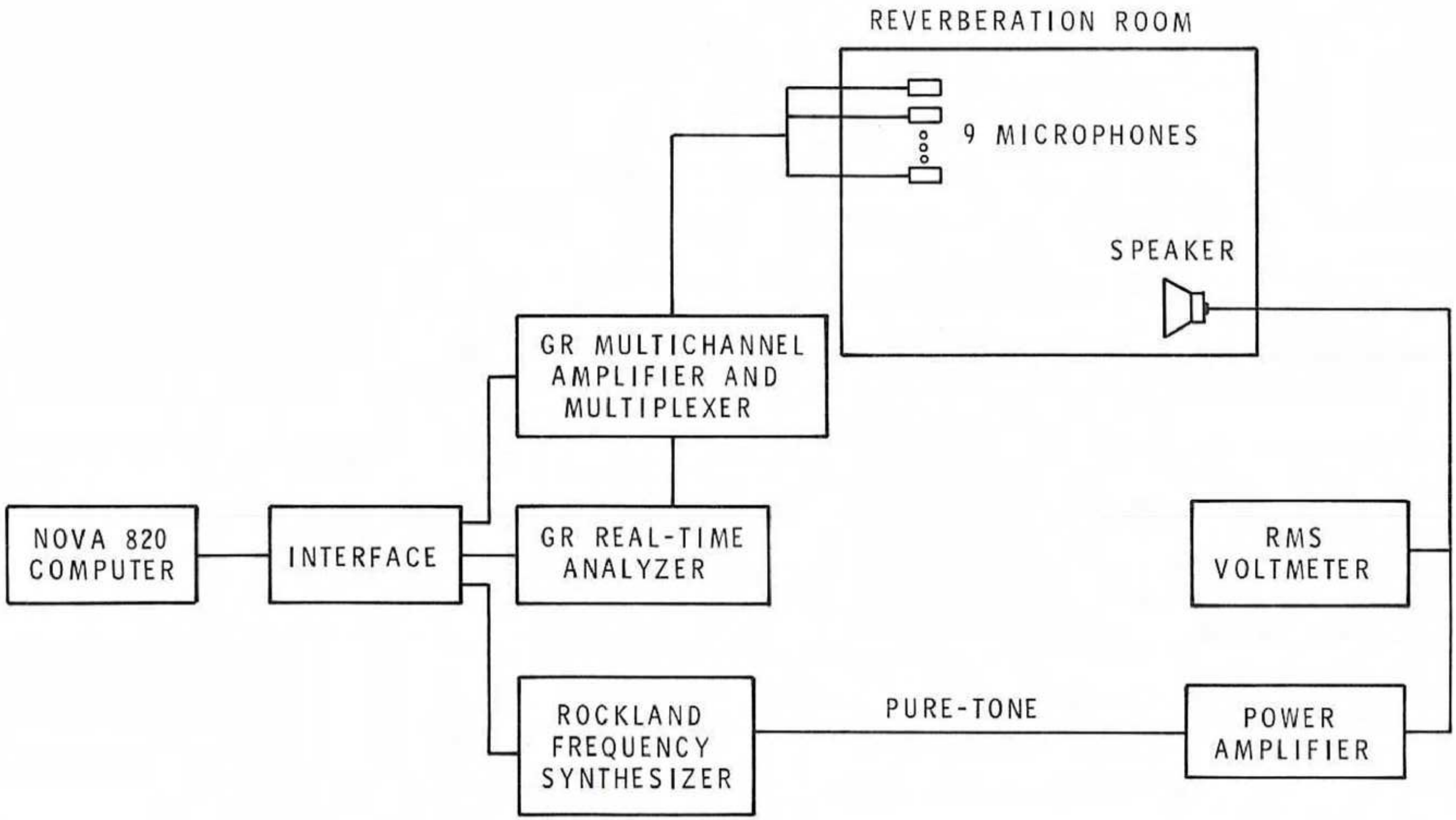
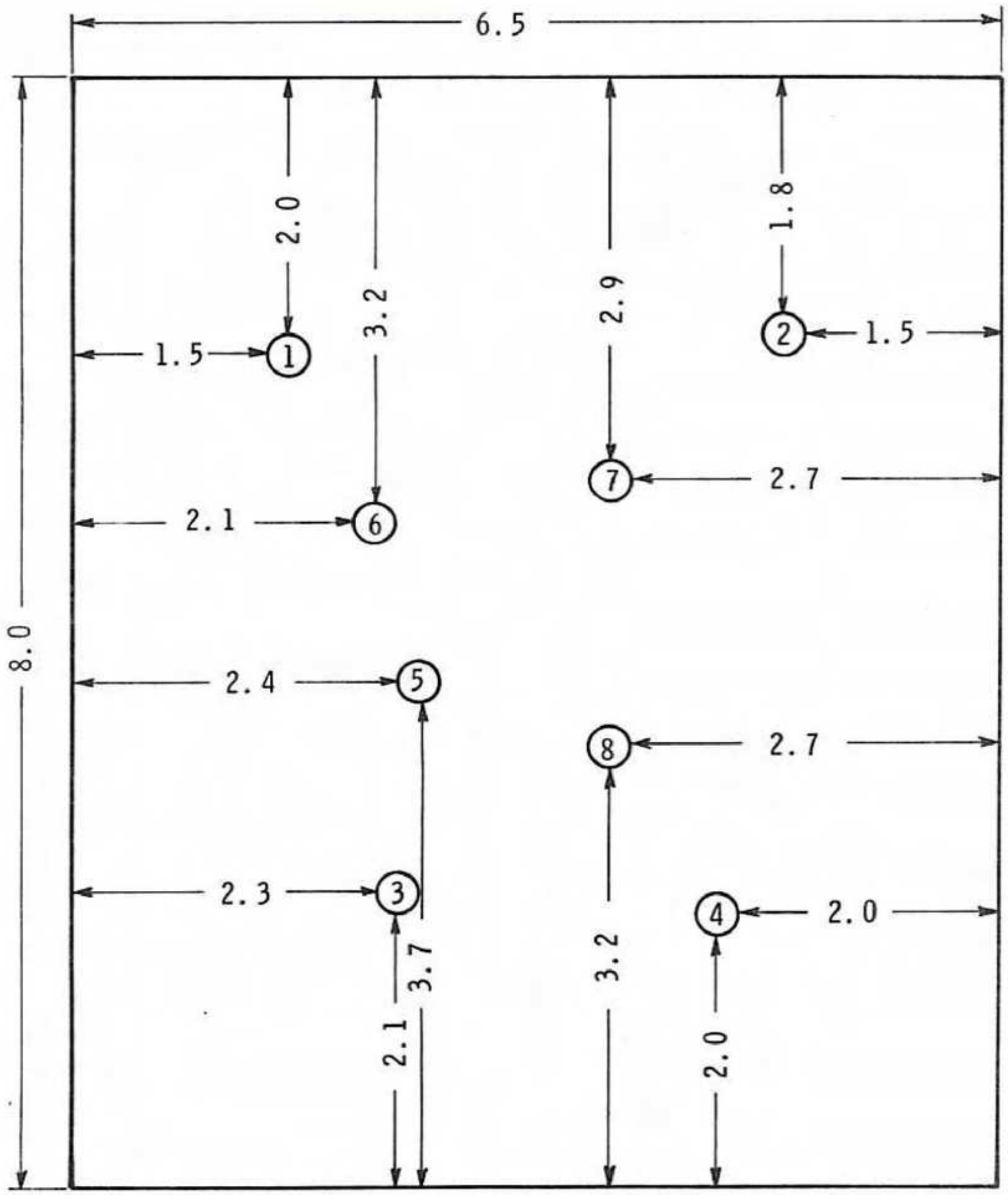


FIGURE 1

BLOCK DIAGRAM SHOWING INSTRUMENTATION SYSTEM USED

BR 6395-1



NOTE: ALL DISTANCES IN METRES

FIGURE 2

SOURCE LOCATIONS ON FLOOR OF THE REVERBERATION ROOM

BR 6395-2

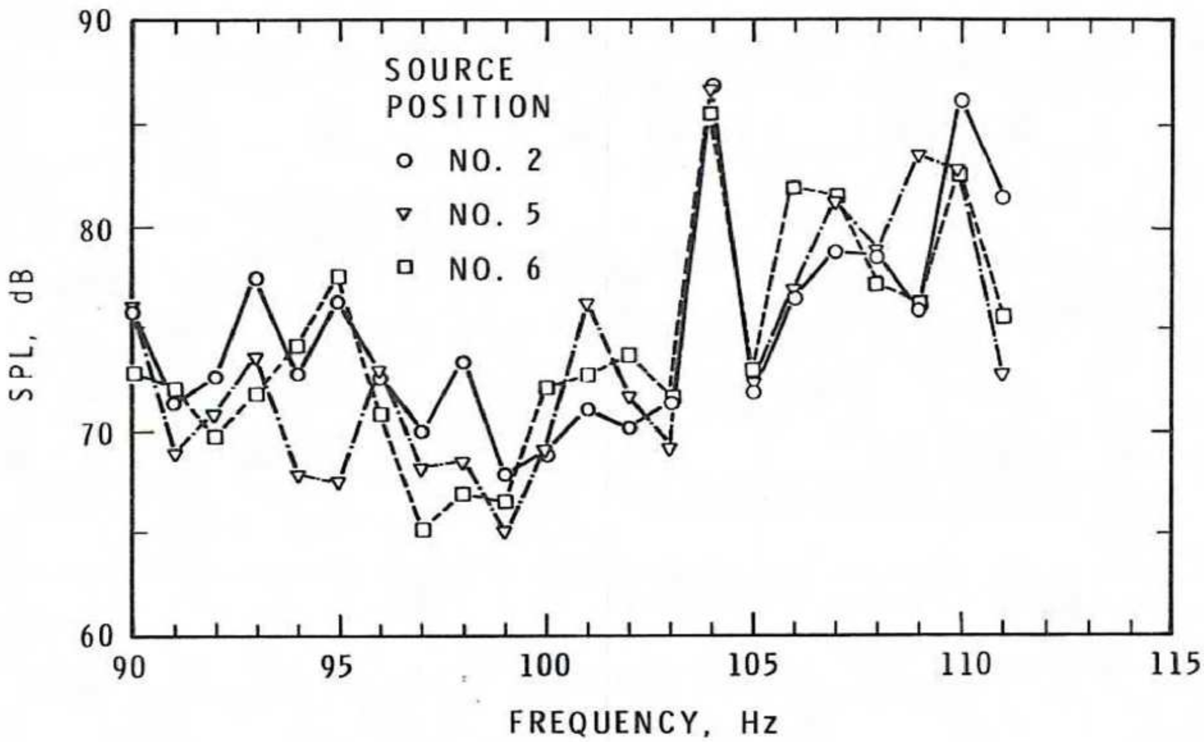


FIGURE 3

SPACE-AVERAGED SOUND PRESSURE LEVELS IN THE REVERBERATION ROOM AS A FUNCTION OF FREQUENCY. TEST FREQUENCIES IN THE 100 Hz, 1/3-OCTAVE BAND, WITH FIXED DIFFUSERS

BR 6395-3

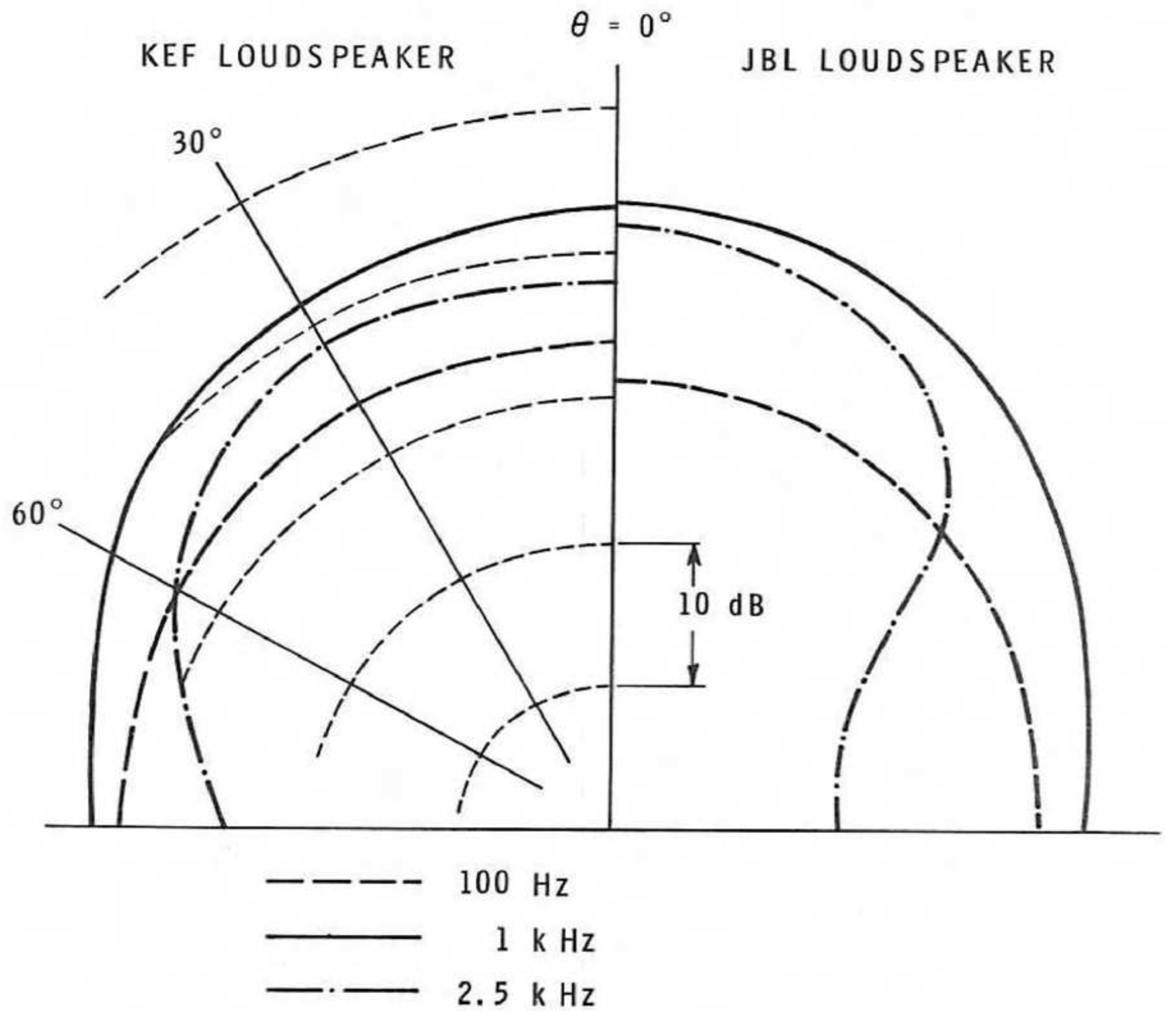


FIGURE 4

DIRECTIVITY PATTERNS OF THE TWO LOUDSPEAKERS
MEASURED IN AN ANECHOIC ROOM AT 1.2 m

BR 6395-4

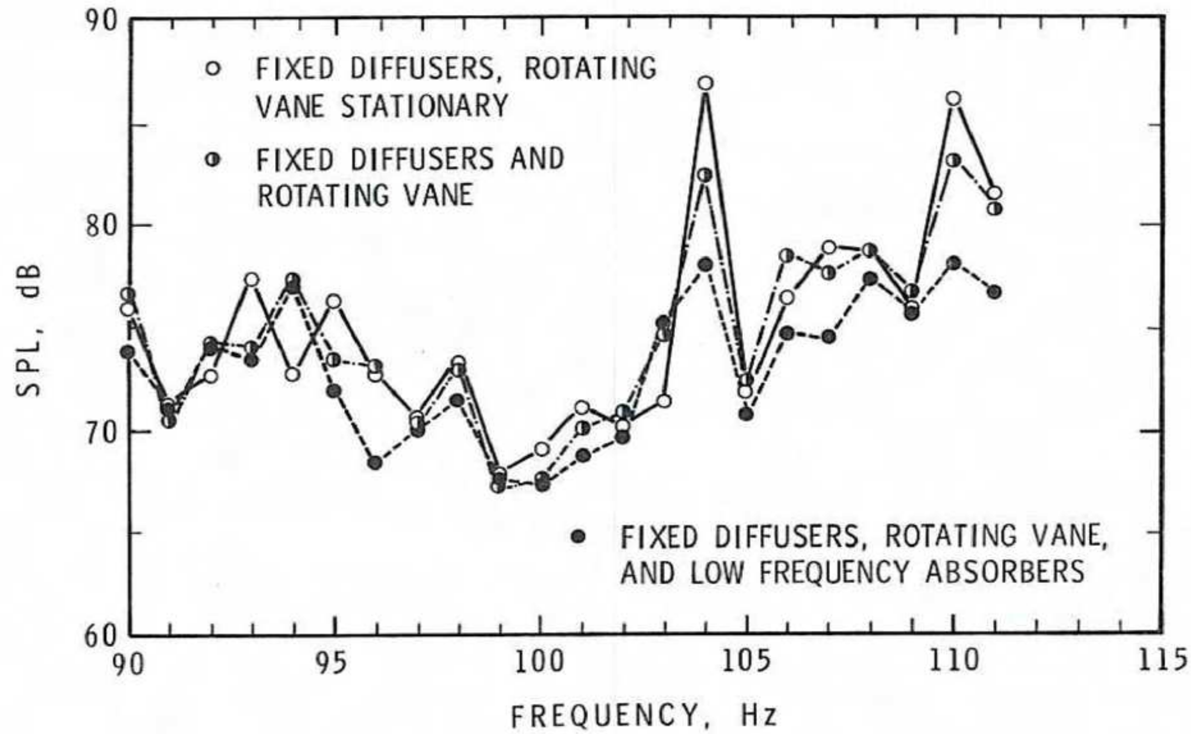


FIGURE 5

SPACE-AVERAGED SOUND PRESSURE LEVELS IN THE REVERBERATION ROOM AS A FUNCTION OF FREQUENCY. TEST FREQUENCIES IN THE 100 Hz, 1/3-OCTAVE BAND, SOURCE AT POSITION NO. 2

BR 6395-5

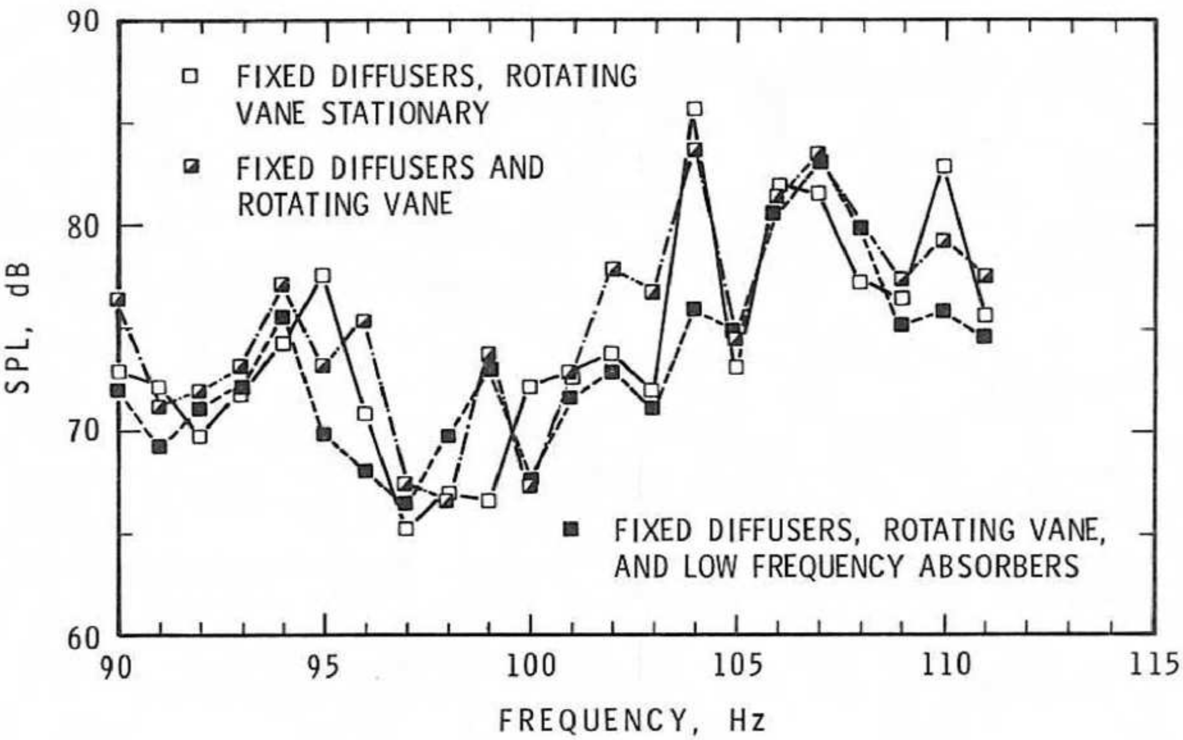


FIGURE 6

SPACE-AVERAGED SOUND PRESSURE LEVELS IN THE REVERBERATION ROOM AS A FUNCTION OF FREQUENCY. TEST FREQUENCIES IN THE 100 Hz, 1/3-OCTAVE BAND, SOURCE AT POSITION NO. 6

BR 6395-6

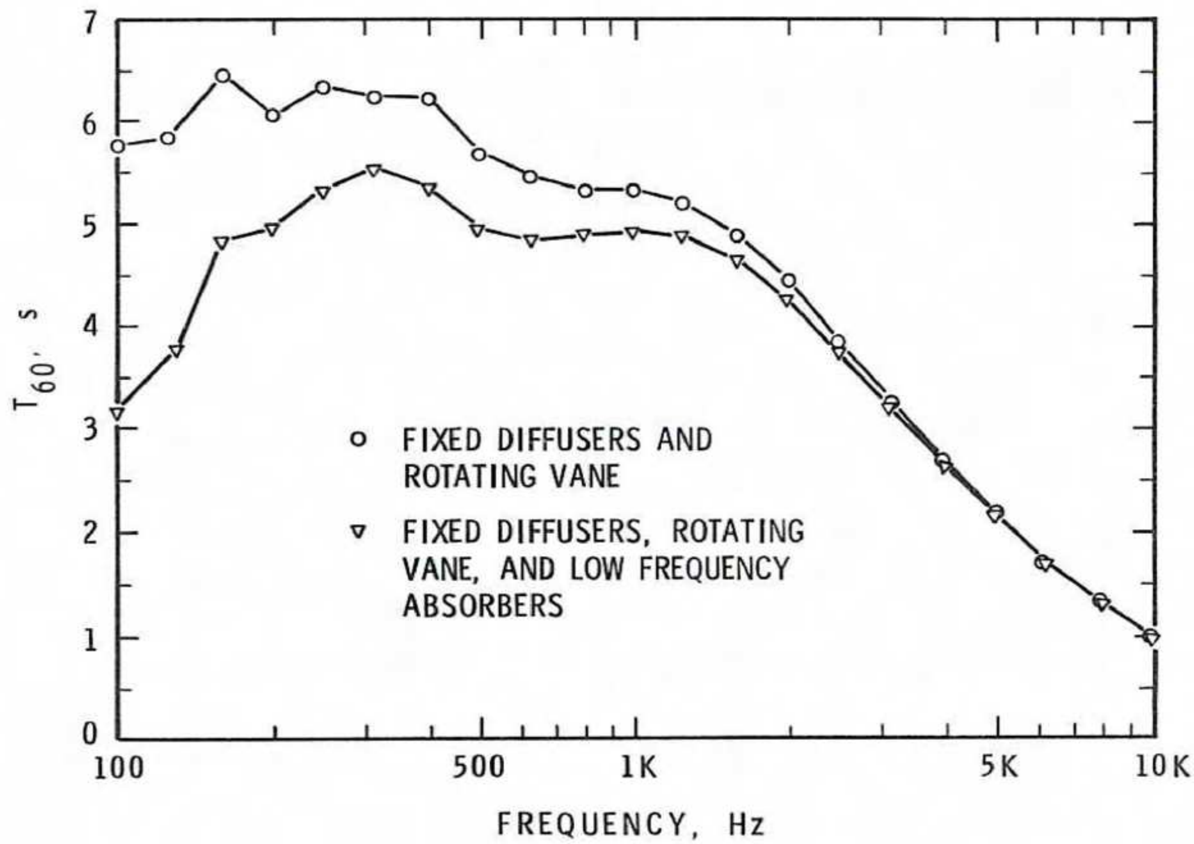


FIGURE 7

REVERBERATION TIME OF ROOM WITH AND WITHOUT LOW FREQUENCY ABSORBERS

BR 6395-7